VLSI Design Course Project
Part I: 4-bit Carry Look-Ahead Adder

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Summary of results

All values below are from ngspice runs, and post-layout values use netlists extracted by MAGIC. Unless stated otherwise, delays are worst-case over all 4608 single-input transitions, measured from the 50% point at the adder input pin to the final 50% crossing at the output. Inputs are driven by a finite inverter, and each output drives the $W_p/W_n = 20\lambda/10\lambda$ inverter.

Table 1: Final design

Quantity	Value
Architecture	Static CMOS prefix CLA with dual-rail AOI22 sum; TSPC flip-flops
Adder transistors / registered transistors	194 / 348
Schematic worst-case delay	516.3 ps
Post-layout worst single-input transition	579.88 ps (a1 rise to s3 rise, A=0001 B=0101 C0=0)
Post-layout worst found overall, tpd,max	589.85 ps (several inputs changing; from a sample)
Course hint vector, post-layout worst	454.1 ps
Flip-flop post-layout tcq / tsetup / thold	204.2 / 38.0 / 45.7 ps
fmax by course formula	1.4847 GHz (T = 673.6 ps)
Verified limit, extracted registered circuit	1.1467 GHz (T = 872.1 ps)
Pavg at verified limit, random and critical vectors	3.3899 mW
PDP (Pavg x tpd,max)	1999.51 fJ
Layout area, adder / registered	2937 / 7006 um2
ADP, adder	1732 um2 ns
DRC errors, adder and registered	0 and 0

Points to note. (1) The course formula gives a higher frequency than the circuit achieves: At the course-formula period (673.6 ps) the extracted registered circuit gave 264 wrong output samples, so that frequency is *not* reliable for this design. The formula leaves out clock-to-Q delay. (2) The hint vector is not the worst case. The slowest arcs start at operand bits, not at C_0. (3) MAGIC has no usable resistance extraction here, so resistance is bounded by simulation in Section 14, not extracted. (4) LVS was run with a script written for this project, because netgen was unavailable. (5) Figure 1(i) shows no register on C_{in}; a ninth input flip-flop was added. (6) The load sentence names sum bits only; C_4 also drives the inverter here.

1 Literature review

Twenty provided papers were read. Few state the output load, input driver or the transitions tested, so their delay figures are not comparable with this work or with each other.

- **OR-type propagate:** Miao and Li [1] replace XOR with NOR in the carry path and name a path from B0 through C3 to S3 as the longest. Both ideas were tested here, and the worst paths here also start at operand bits.
- **Mirror and flat carry gates:** Hasan et al. (2019) build C_4 as one nested gate. Measured stacks here show why this is costly: a worst-position NOR5 is 4.6 FO4 at 180 nm.
- **Reduced-swing styles:** GDI and pass-transistor papers (Dipto 2020, Hossain 2020, Joy 2023, Chezhiyan 2025) report gains against their own baselines at 90 nm. One of these (Chezhiyan and Bommi 2025) itself lists static CMOS as fastest. They were not adopted.
- **Dynamic logic:** Zlatanovici et al. [4] show dynamic logic winning at 64 bits, and static preferred at long cycle times. Efstathiou et al. (2013) report only 4.73% gain at 8 bits from a Ling-style split. Neither applies directly at 4 bits.
- **False sums:** Nyathi and Mutumba (2017) show sums switching before carries settle, which is why delay here is measured at the final crossing.
- **Single-threshold limit:** MT-CMOS and dual-mode designs (Sivakumar 2013, Chaitanya Kumar 2014) need low-threshold devices that the provided model does not offer.

Full notes, with source pages, are in `explanation/research/literature_review.md`.

2 Proposed structure

2.1 Equations and structure

With $p_i = a_i \oplus b_i$, $g_i = a_i b_i$, $t_i = a_i + b_i$, the carries can use t_i in place of p_i, because the two differ only when $g_i = 1$. XOR is then needed only for the sum, $S_i = p_i \oplus c_i$, so the carry path is free of a slow XOR. This idea follows Miao and Li [1].

The carries are computed as a prefix tree with C_0 entering last:

$$G_{1:0} = g_1 + t_1 g_0, \quad T_{1:0} = t_1 t_0, \quad G_{3:2} = g_3 + t_3 g_2, \quad T_{3:2} = t_3 t_2,$$

$$G_{3:0} = G_{3:2} + T_{3:2} G_{1:0}, \quad G_{2:0} = g_2 + t_2 G_{1:0}, \quad C_k = G_{k-1:0} + T_{k-1:0} C_0.$$

Each level is inverting, so polarity alternates: NAND2 and NOR2 give active-low $\bar{g}, \bar{t}$; OAI21 and NOR2 give active-high level 1; AOI21 and NAND2 give active-low level 2; a final OAI21 with $\bar{C}_0$ gives each carry. C_0 passes through one inverter and one gate.

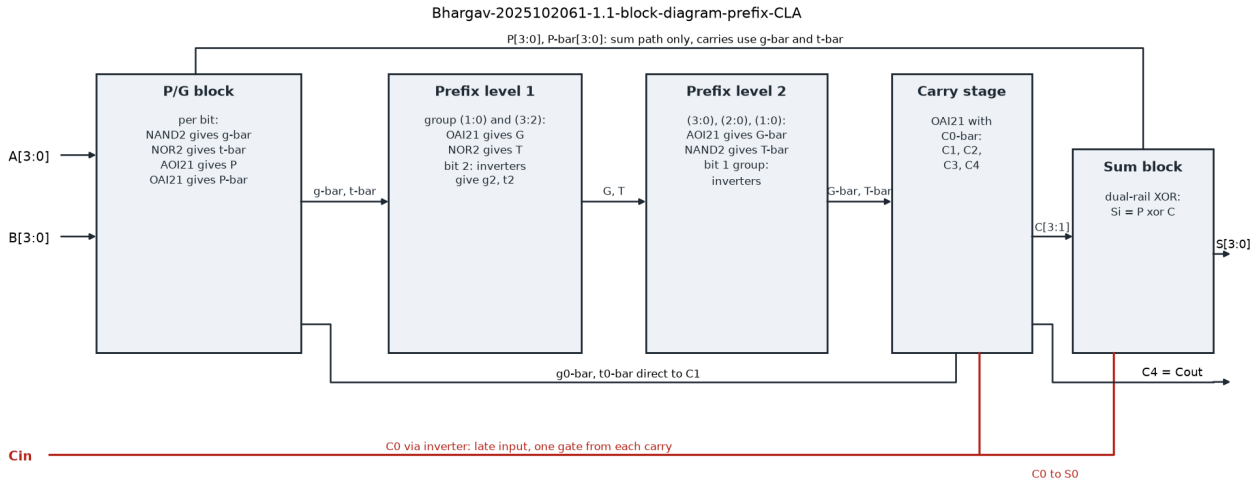


Figure 1: Block-level architecture. Signals are named as in the netlists.

2.2 How this architecture was chosen

Six architectures were simulated at unit sizing, over all 4608 single-input transitions. Delay from the driver input also charges each design for its input capacitance.

Table 2: Unit-size comparison, schematic, all single-input transitions

Candidate	T	Width (lambda)	Worst pin (ps)	From driver (ps)	Hint vector (ps)
C1	208	5260	771.3	832.5	421.4
C2	170	5470	764.0	840.6	577.2
R	172	5160	727.0	803.5	574.2
C3	170	4800	796.4	873.1	623.5
C4	194	4930	658.9	735.5	360.0
E	144	3480	799.8	882.2	629.1
R/tg	164	4560	698.8	787.0	542.1
R/static12	180	5280	726.8	803.4	574.2
R/nand4	196	5360	740.4	848.8	605.5

C1 is the textbook XOR and NAND-NAND design; C2 uses one complex gate per carry, up to a 5-high stack; R is C2 with C_4 from a mirror gate over C_2; C3 limits stacks to 3; C4 is the prefix tree; E is a static Manchester chain. A first-order logical-effort estimate had placed flat designs ahead of the prefix tree. Measurement contradicted it, so the estimate was dropped. The finalists were then sized with the same method and laid out:

Table 3: Finalists after sizing, laid out and extracted. Power is the adder alone on the same 200 random vectors at 1 ns per vector.

Design	T	Post, sized on schematic (ps)	Post, final, single-input (ps)	Area (um2)	ADP (um2 ns)	P (mW)	PDP (fJ)
C4 prefix, AOI22 sum (chosen)	194	597.6	579.9	2937	1703	1.2704	737
C4 prefix, TG sum	186	601.5	596.7	2498	1491	1.0524	628
R flat carries, TG sum	164	-	645.6	3426	2212	0.8505	549

The AOI22 sum version is fastest, and speed is the primary objective, so it was chosen. The TG sum version is 16.8 ps slower with lower area and ADP, so it is the better choice if area is weighted more. R with AOI22 sums reached 586.1 ps on schematic, but its assembled layout failed LVS after routing relaxation, so it has no valid post-layout number and is not listed. C2 was stopped early at 663.5 ps on schematic, well behind C4.

3 Design details: topology and sizing

3.1 Adder

Sizes come from an automated search minimising worst-case delay from the driver input, confirmed against every input-output class, then repeated with wire capacitance extracted from the layout. Each accepted step is logged in `part1/simulations/optimisation/`. Widths below are in λ ($\lambda = 0.09\,\mu$m); every channel length is 2λ, equal for NMOS and PMOS. Multiple widths are the different series positions in a stack.

Table 4: Final adder sizing, every instance

Instance	Cell	Output	T	NMOS W	PMOS W
nand0	nand2	g0b	4	25	25
nor0	nor2	t0b	4	16	62
aoi0	aoi21	p0	6	4/8	16
oai0	oai21	p0b	6	8	8/16
nand1	nand2	g1b	4	25	25
nor1	nor2	t1b	4	20	78
aoi1	aoi21	p1	6	4/8	16
oai1	oai21	p1b	6	8	8/16
nand2	nand2	g2b	4	25	25
nor2	nor2	t2b	4	12	50
aoi2	aoi21	p2	6	8/16	32
oai2	oai21	p2b	6	16	16/32
nand3	nand2	g3b	4	20	20
nor3	nor2	t3b	4	12	50
aoi3	aoi21	p3	6	10/20	40
oai3	oai21	p3b	6	20	20/40
invc0	inv	c0b	2	10	20
g10	oai21_fn	g1_0	6	25	25/50
t10	nor2	t1_0	4	16	62
g32	oai21	g3_2	6	16	16/32
t32	nor2	t3_2	4	8	32
ig2	inv	g2	2	12	25
it2	inv	t2	2	20	39
g30	aoi21_fnp	g3_0b	6	4/8	16
t30	nand2	t3_0b	4	8	8
g20	aoi21_fnp	g2_0b	6	10/20	40
t20	nand2	t2_0b	4	25	25
ig10	inv	g1_0b	2	10	20
it10	inv	t1_0b	2	16	31
c1	oai21	c1	6	10	10/20
c2	oai21_fn	c2	6	25	25/50
c3	oai21_fn	c3	6	25	25/50
c4g	oai21_fnp	c4	6	16	16/32
x0	xor_aoi22	s0	10	5/25	10/50
x1	xor_aoi22	s1	10	6/20	13/40
x2	xor_aoi22	s2	10	12/16	25/32
x3	xor_aoi22	s3	10	10	20

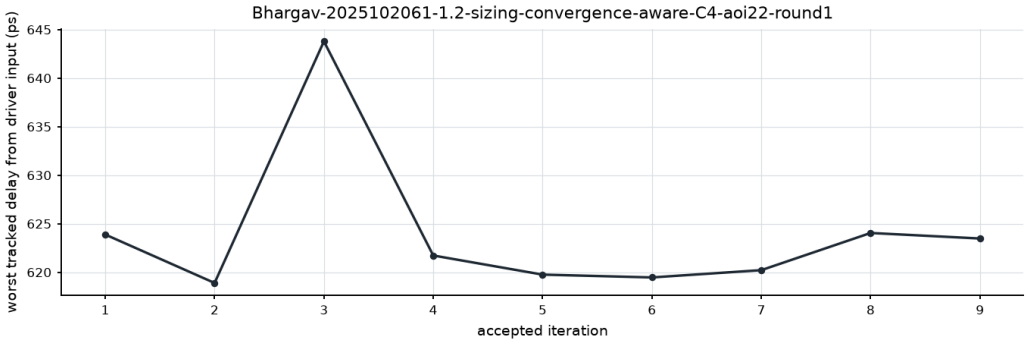


Figure 2: Layout-aware sizing, first round. Each point is an accepted step.

3.2 Transistor-level schematics

Each drawing is generated from the same definition that writes the netlist, and a test confirms each drawing shows exactly the netlist's devices, terminals and sizes. All final cells are in `part1/schematics/transistor_level/final/`. Representative cells follow.

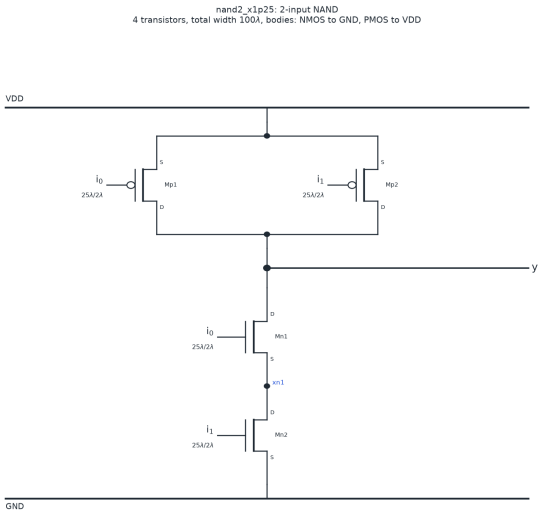


Figure 3: NAND2 for generate, bit 0 (nand2_x1p25).

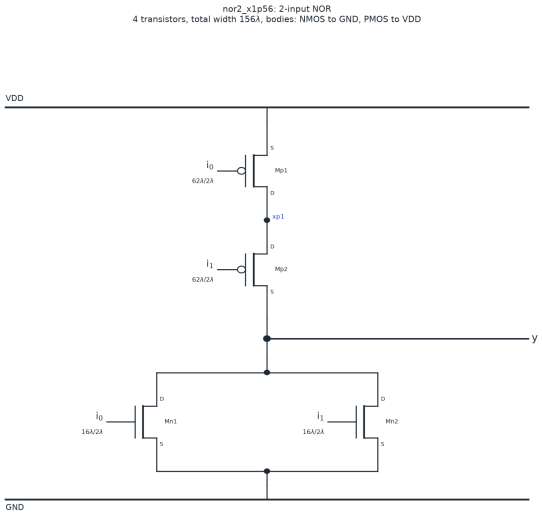


Figure 4: NOR2 for OR-type propagate, bit 0 (nor2_x1p56).

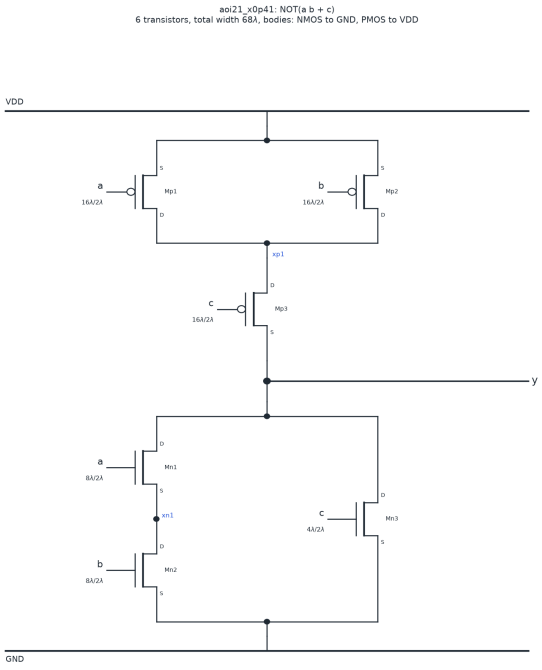


Figure 5: AOI21 for P, bit 0 (aoi21_x0p41).

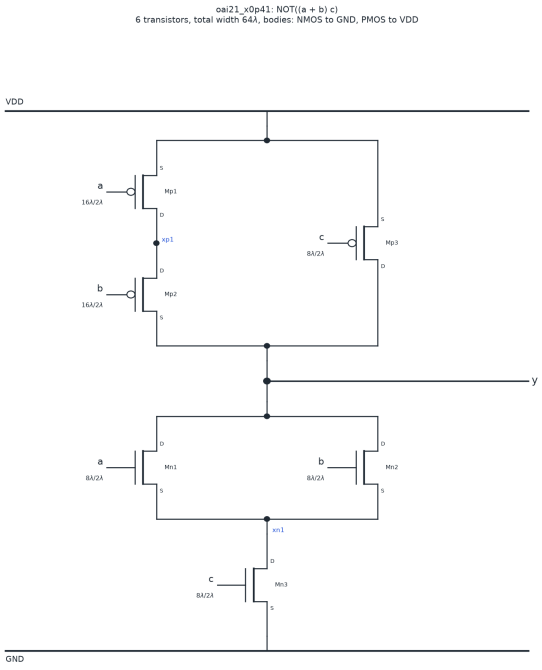


Figure 6: OAI21 for P-bar, bit 0 (oai21_x0p41).

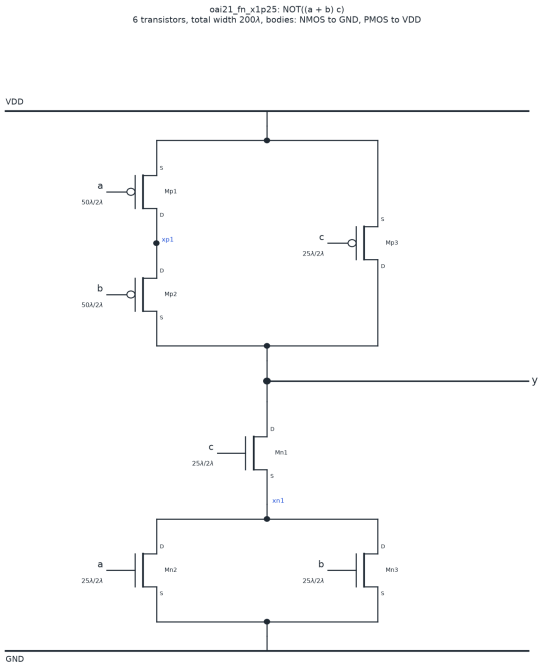


Figure 7: Prefix OAI21, G(1:0) (oai21_fn_x1p25).

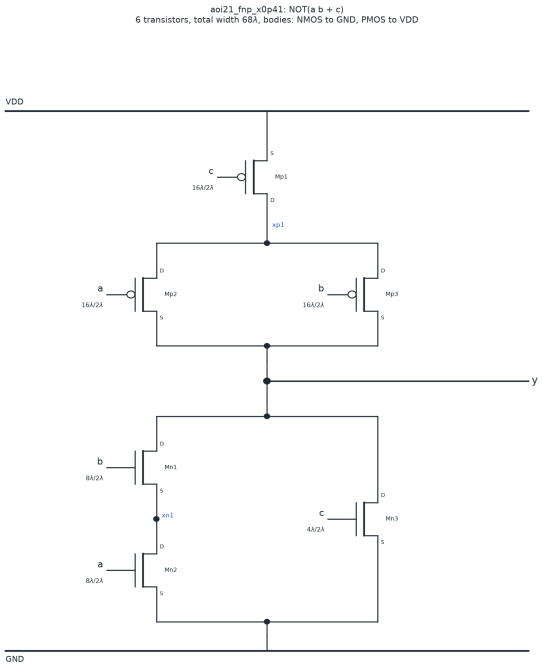


Figure 8: Prefix AOI21, G-bar(3:0) (aoi21_fnp_x0p41).

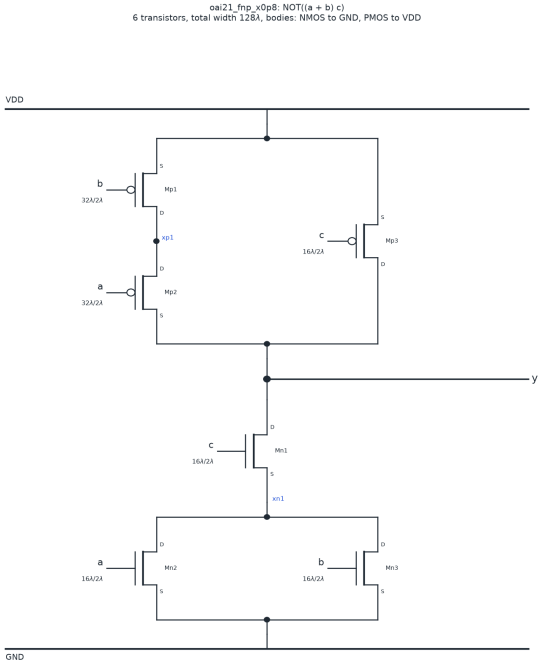


Figure 9: Final carry OAI21, C4 (oai21_fnp_x0p8).

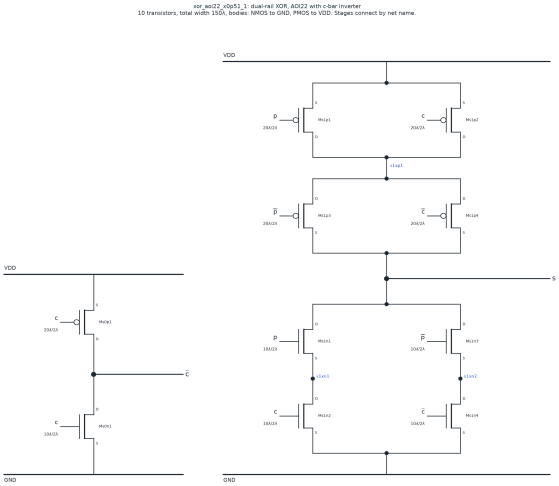


Figure 10: Dual-rail XOR for S3 with c-bar inverter (xor_aoi22_x0p51_1).

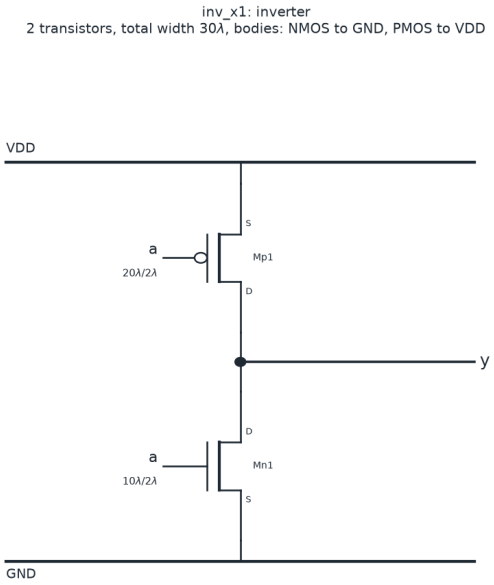


Figure 11: C0 inverter (inv_x1).

3.3 D flip-flop

TSPC and a static transmission-gate master-slave flip-flop were compared with a 15 fF load: TSPC 172.9 ps clock-to-Q and 40.0 ps setup; master-slave 200.8 ps and 116.0 ps. TSPC is about 100 ps better on clock-to-Q plus setup, with 11 transistors against 22. The course formula alone slightly favours the master-slave because of its negative hold time, but it would give a slower clock in practice, so TSPC was chosen. TSPC nodes are dynamic, so it needs a minimum clock frequency; this was not characterised.

Stage drives are $\times 0.6$, $\times 1.0$, $\times 1.0$, $\times 2.0$, found by search at the largest adder pin load, 47.8 fF including wire.

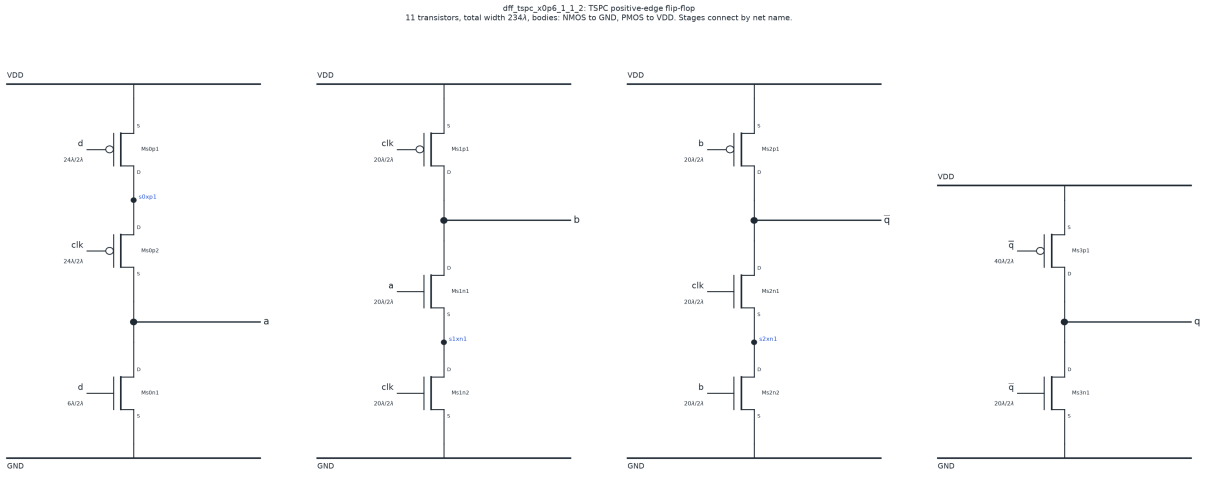


Figure 12: Final TSPC flip-flop.

4 Block simulation

Every cell and block was exhaustively simulated at transistor level through finite inverter drivers, with the $20\lambda/10\lambda$ inverter on each output, in binary and in shuffled order. A DC solution for every vector checks for static current. The unit-size 5-high C_4 gate, used only by the rejected candidate C2, needed 3 ns per vector to settle within 10% of VDD (1.604 V at 980 ps), so its row uses 3 ns; all others use 1 ns.

Table 5: Truth-table results, library at unit size

Cells and blocks	Failing	Worst VOH (V)	Worst VOL (V)	Largest DC current (nA)
30	0	1.7289	0.0113	0.72

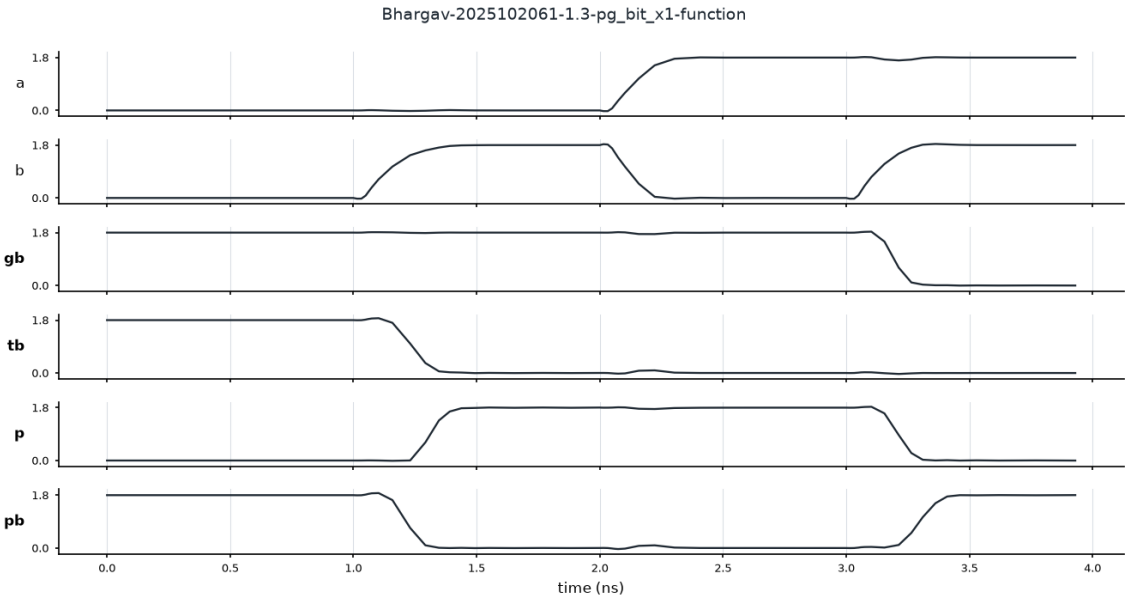


Figure 13: P/G bit waveforms.

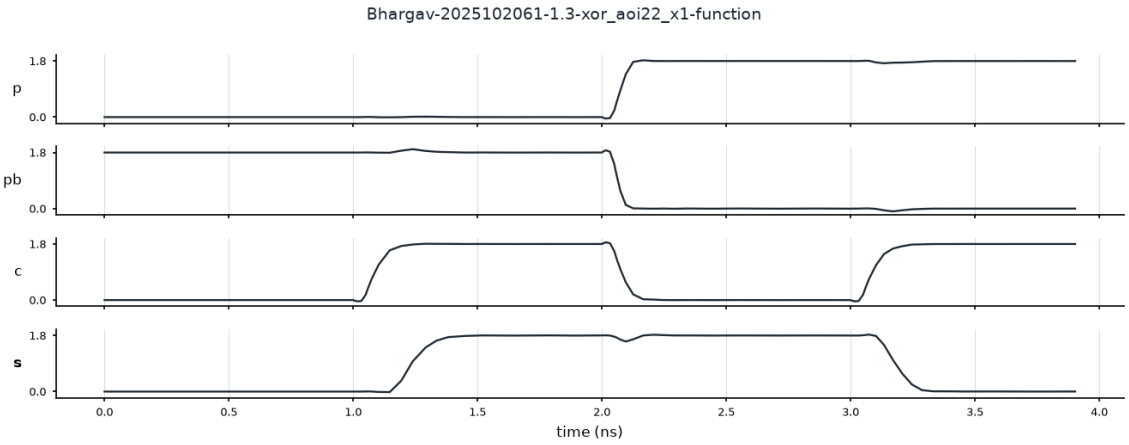


Figure 14: Dual-rail XOR waveforms.

Final adder, all 512 input combinations, each twice: schematic 0 errors, extracted 0 errors.

5 Flip-flop characterisation

Measured at 50% VDD. t_{cq} is with data well before the clock. t_{setup} is the shortest D-to-clock time keeping clock-to-Q within 10% of t_{cq}. t_{hold} is the shortest clock-to-D time for which a later data change leaves Q correct, with no disturbance above 10% of VDD. Both data values were tested and the larger is reported. Definitions follow Stojanović and Oklobdžija [2].

Table 6: Flip-flop timing at 47.8 fF load

Quantity (ps)	Schematic	Post-layout
Clock to Q, Q rising	134.1	136.1
Clock to Q, Q falling	208.5	204.2
Setup, D rising	-26.0	-8.0
Setup, D falling	34.0	38.0
Hold, D rising	-11.7	-11.7
Hold, D falling	55.1	45.7
Worst tcq / tsetup / thold	208.5 / 34.0 / 55.1	204.2 / 38.0 / 45.7

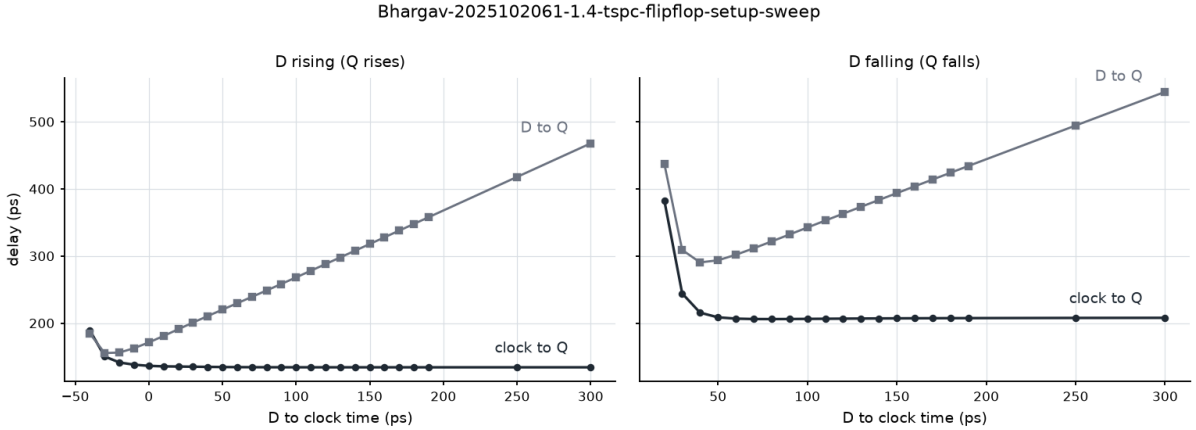


Figure 15: Delay against D-to-clock time, schematic.

6 Block layouts

Layouts were generated by script for the provided technology file and checked with MAGIC. Every cell passes DRC with zero errors, and extraction is compared with the schematic by the project's LVS script, which merges parallel fingers and is tested against wrong widths, swapped gates, missing devices and shorted nets.

Table 7: Each block: schematic against extracted, worst over all its single-input transitions

Block	T	W x H (lambda)	Area (um2)	DRC	LVS	Sch tpLH	Sch tpHL	Post tpLH	Post tpHL
pg0	20	137 x 178	197.5	0	match	187.21	168.95	201.02	176.08
pg1	20	137 x 194	215.3	0	match	187.4	171.1	200.97	169.7
pg2	20	137 x 166	184.2	0	match	195.19	176.14	206.87	180.24
pg3	20	137 x 161	178.7	0	match	199.86	189.09	210.24	190.19
carry	74	525 x 223	948.3	0	match	319.87	306.42	335.79	336.37
sum0	10	84 x 166	112.9	0	match	176.33	173.69	182.49	171.72
sum1	10	84 x 151	102.7	0	match	157.14	144.1	161.33	142.78
sum2	10	84 x 139	94.6	0	match	134.23	119.06	134.85	118.7
sum3	10	84 x 121	82.3	0	match	134.71	114.42	137.22	117.54

Table 8: Flip-flop layout

Width x height (lambda)	Area (um2)	DRC	LVS
99 x 142	113.9	0	match

The layout figures below are screenshots of MAGIC's own layout window, taken with the cell loaded and fitted to the window.

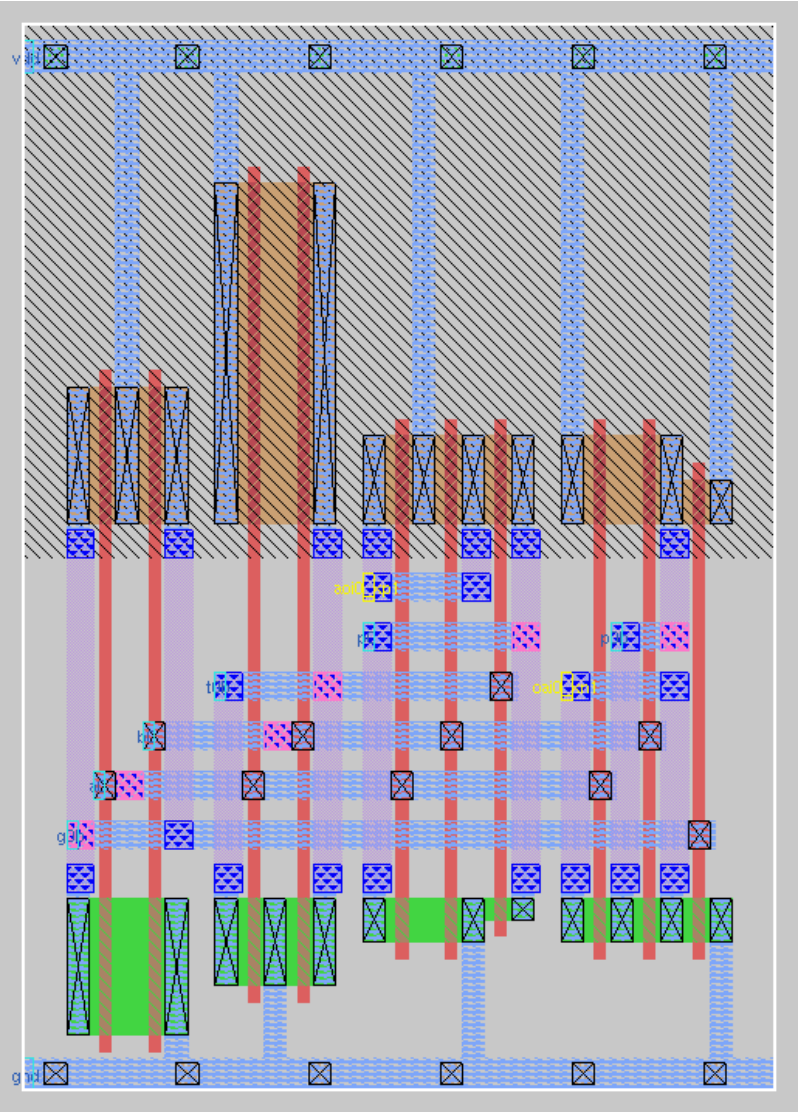


Figure 16: P/G bit 0 tile in MAGIC, as placed in the final adder.

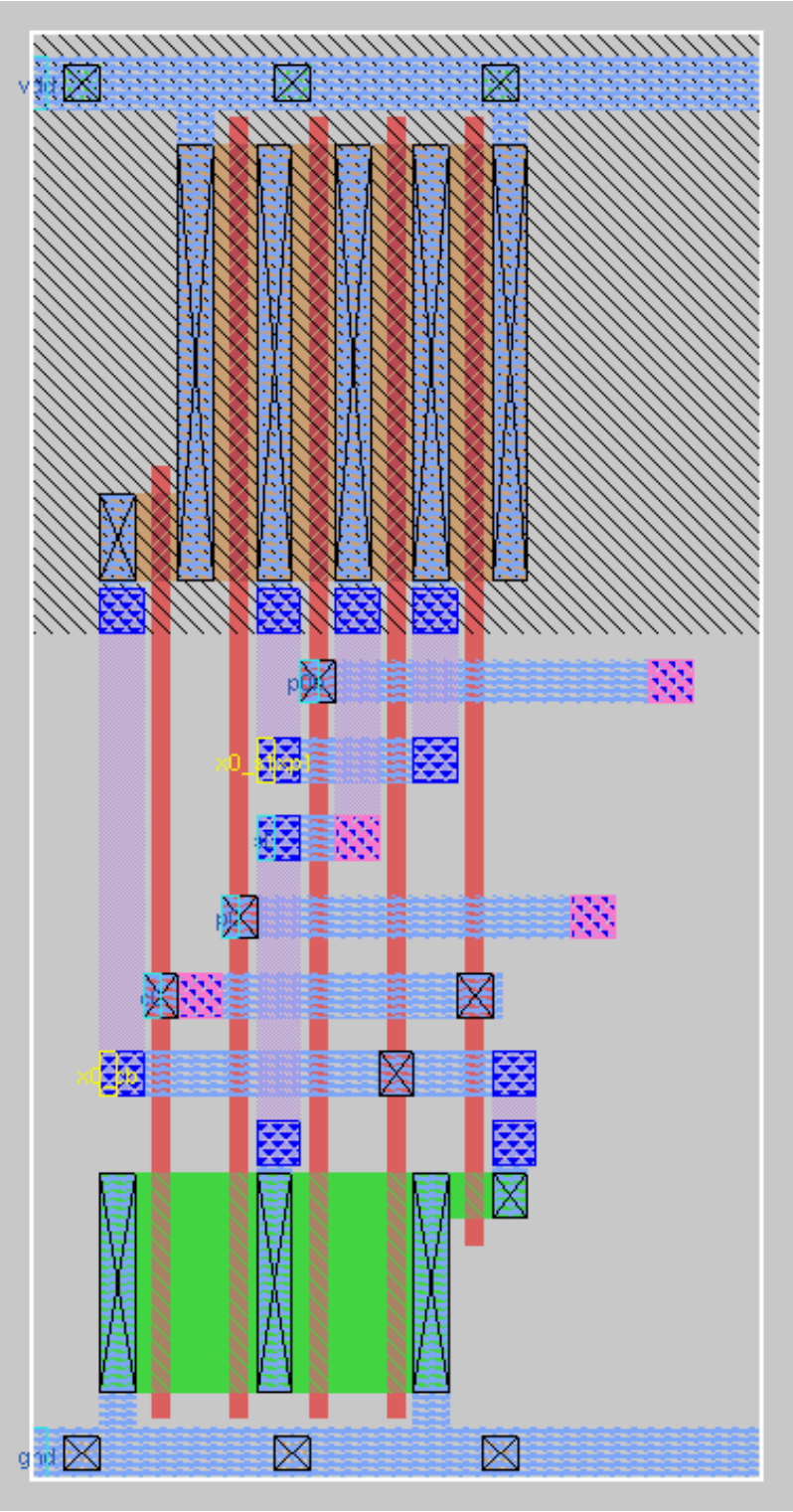


Figure 17: Sum bit 0 tile in MAGIC.

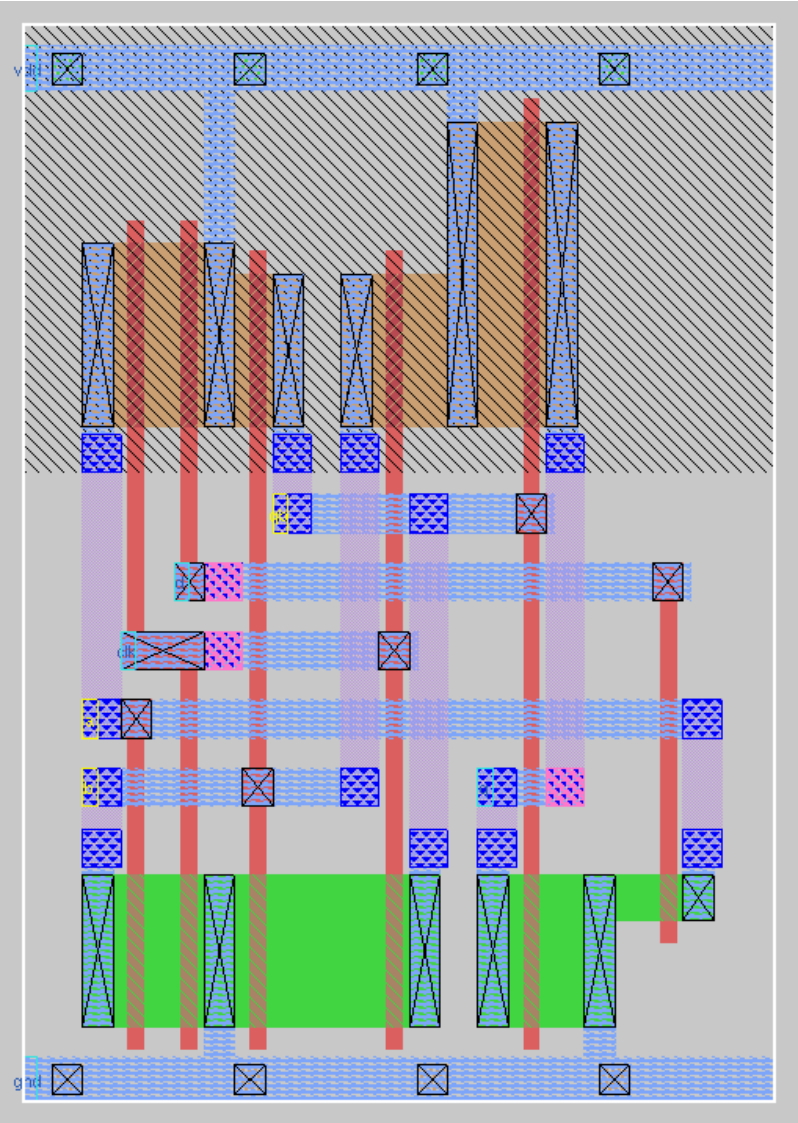


Figure 18: TSPC flip-flop layout in MAGIC.

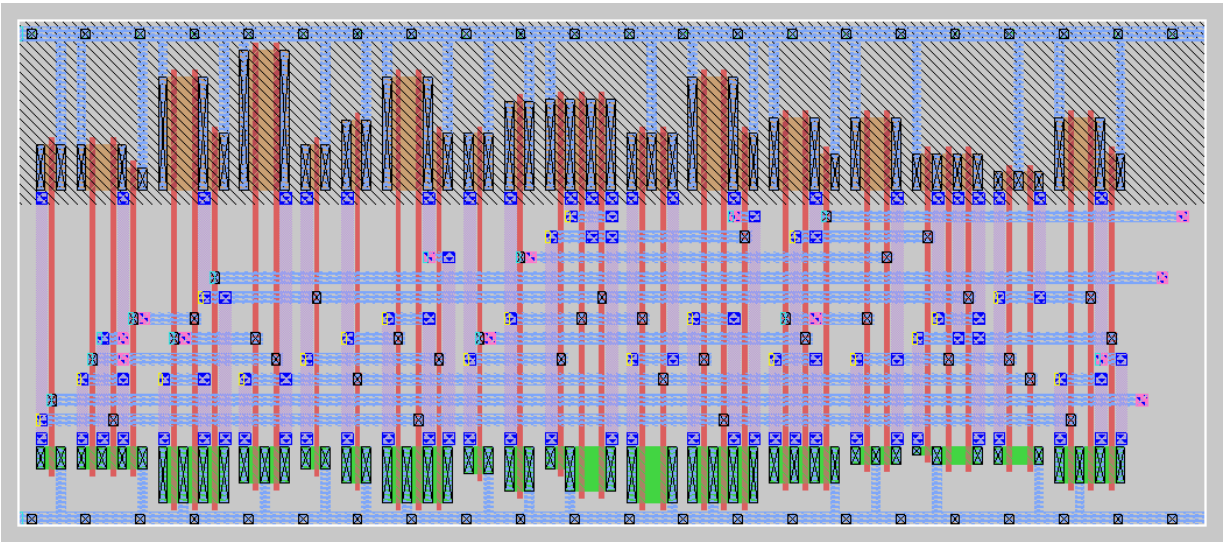


Figure 19: Carry block tile in MAGIC.

7 Integration

The complete circuit of Figure 1(i) has 9 input flip-flops, the adder and 5 output flip-flops. It was simulated with the hint vector, the 30 slowest arcs from the adder search, and 200 random vectors. A vector captured at one rising edge must appear at the output flip-flops after the next. The limit below is the shortest clock period, found by binary search to 5 ps, with zero wrong samples.

Table 9: Registered system

Quantity	Schematic	Post-layout
Vectors checked	264	264
Shortest working period (ps)	740.2	872.1
Maximum verified frequency (GHz)	1.3509	1.1467
Wrong samples at that period	0	0

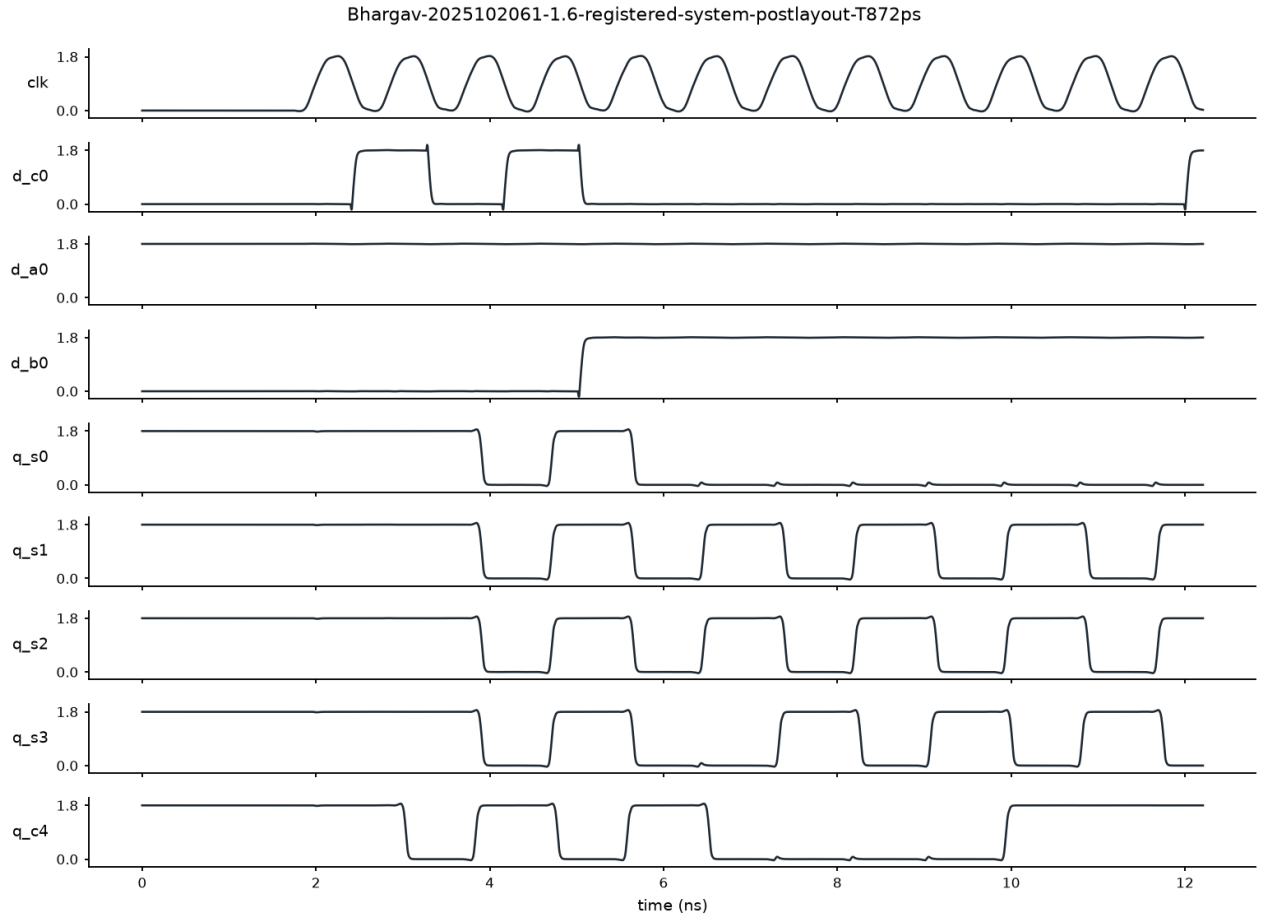


Figure 20: Extracted registered circuit at the verified limit, first vectors. Q values follow inputs after one clock period.

8 Floor plan and pitches

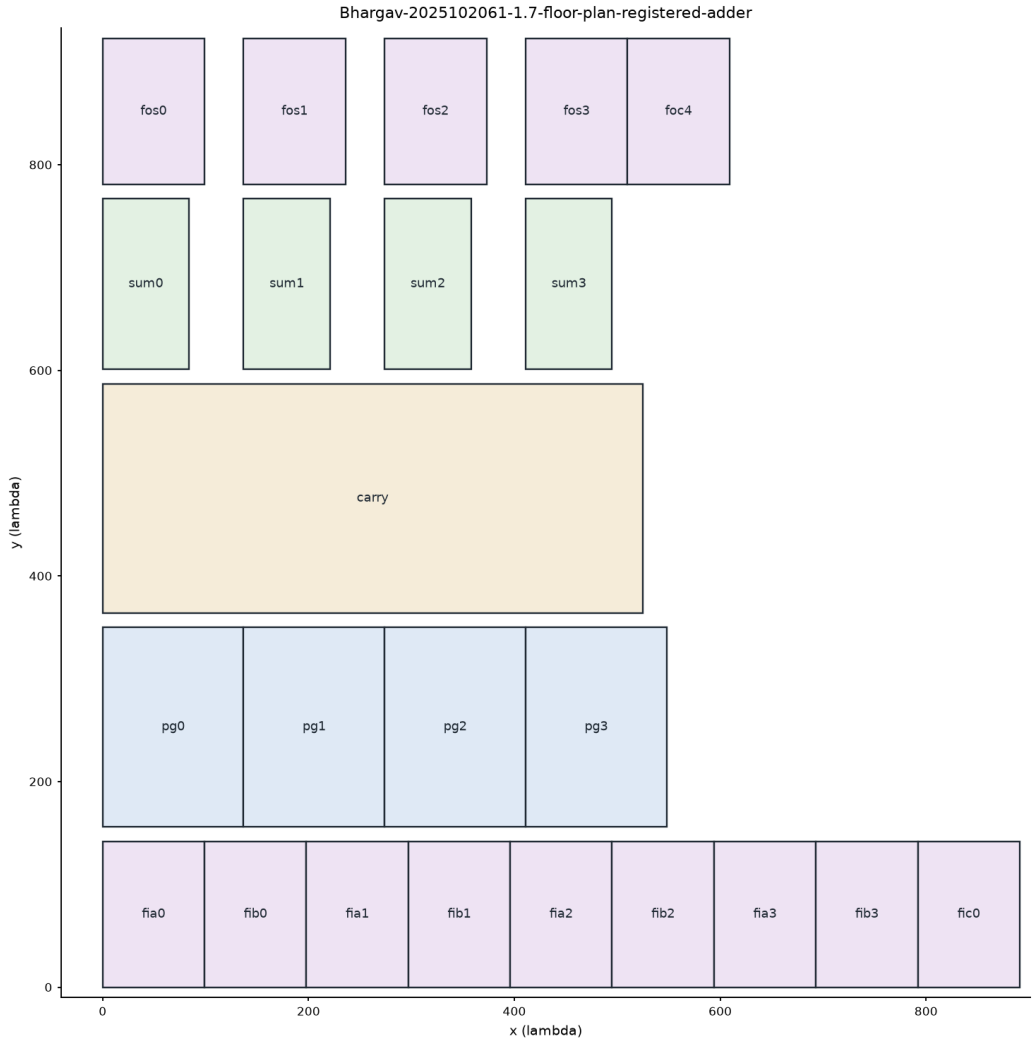


Figure 21: Floor plan of the registered adder. Rows from bottom: input flip-flops, P/G bits, carry block, sum bits, output flip-flops.

The P/G bit cells form the regular structure, and sum cells align to their columns.

Table 10: Pitches

Structure	**Pitch (lambda)**	**Pitch (um)**
Horizontal: bit pitch, P/G and sum columns	137	12.33
Horizontal: flip-flop cell	99	8.91
Vertical: row pitches, bottom to top	156, 208, 237, 180	14.04, 18.72, 21.33, 16.20
Channel track pitch, metal 1	9	0.81
Gate pitch, shared / contacted diffusion	7 / 9	0.63 / 0.81
Over-cell routing grid, metal 3 and 4	9	0.81

9 Full layout: schematic against post-layout

Adder layout: DRC 0, LVS match: 194 merged devices, 108 nets, 712 extracted capacitors totalling 376.9 fF. Registered layout: DRC 0, LVS match: 348 merged devices, 207 nets, area 7005.5 μm^2.

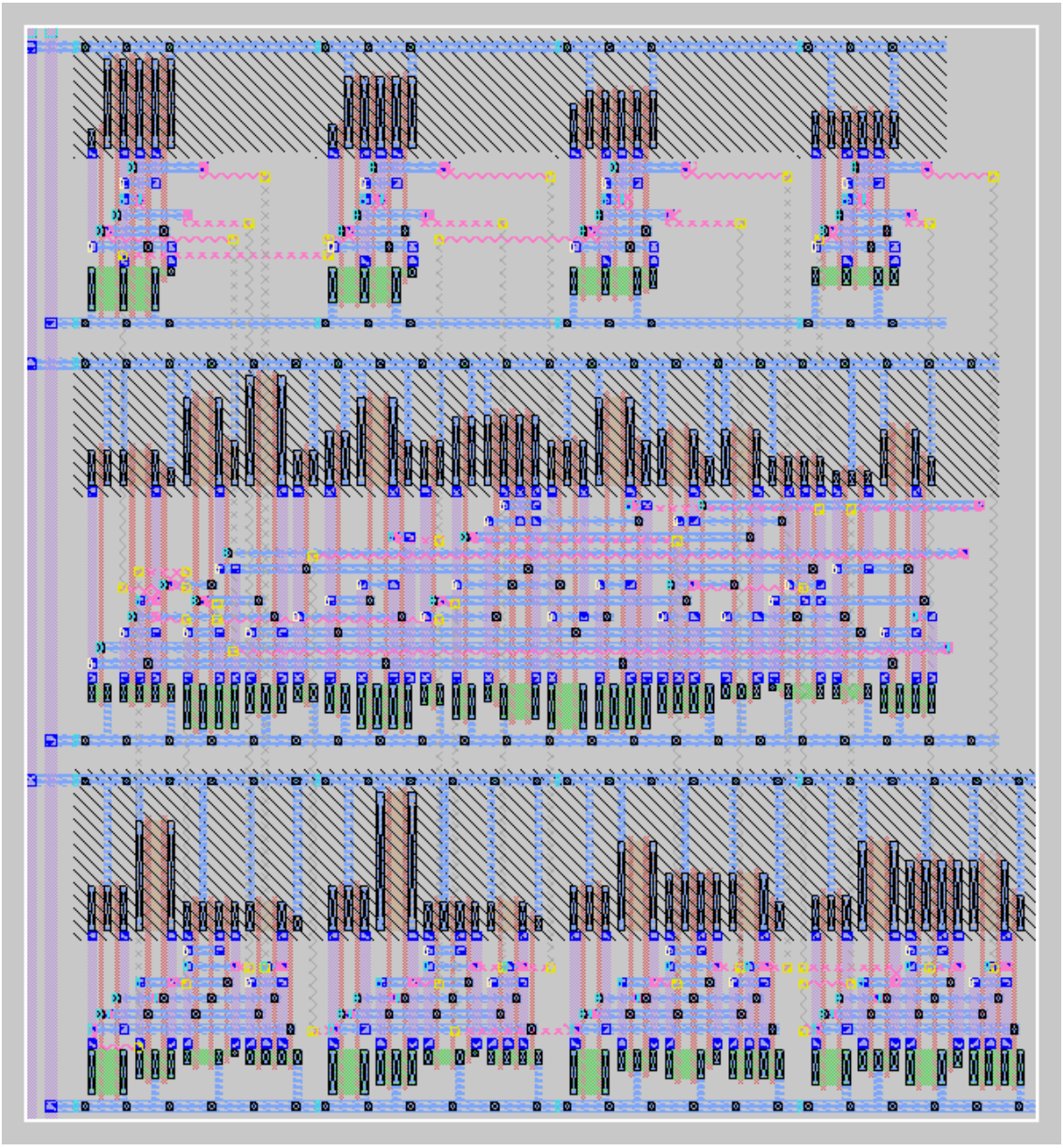


Figure 22: Final 4-bit adder layout in MAGIC: P/G row at the bottom, carry block, sum row at the top.

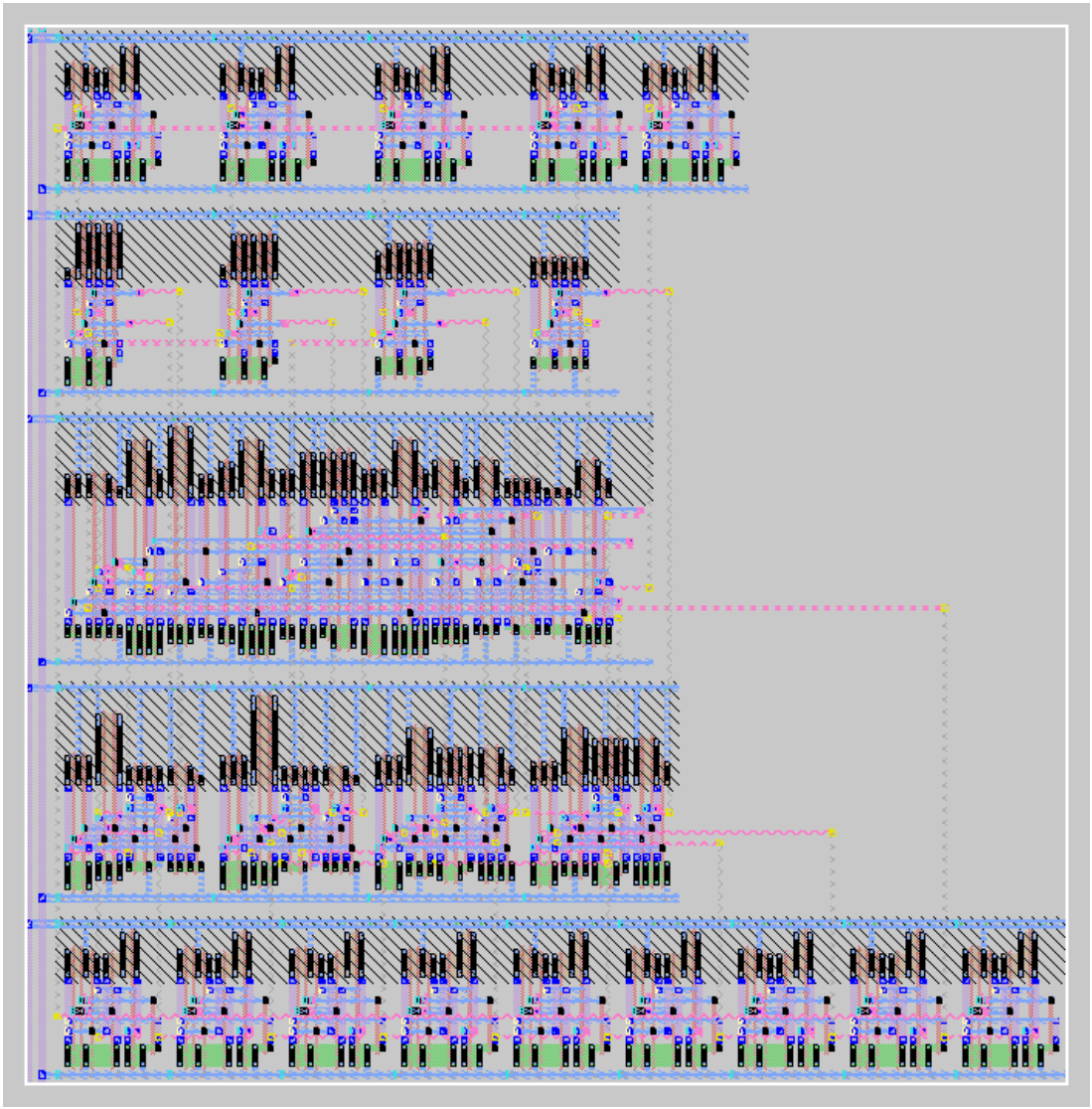


Figure 23: Registered adder layout in MAGIC: input flip-flops at the bottom, output flip-flops at the top.

Table 11: Worst delay per output, all single-input transitions (ps)

Output	Edge	Schematic	Post-layout	Change (%)
S0	tpLH	349.7	387.1	10.7
S0	tpHL	352.7	394.8	11.9
S1	tpLH	429.9	536.9	24.9
S1	tpHL	409.2	508.1	24.2
S2	tpLH	499.7	562.8	12.6
S2	tpHL	470.1	543.6	15.6
S3	tpLH	516.3	579.9	12.3
S3	tpHL	491.1	570.2	16.1
C4	tpLH	414.2	490.4	18.4
C4	tpHL	399.2	466.4	16.9

Worst single-input delay rose from 516.3 to 579.9 ps (12.3%). Note that the final sizes were tuned with wire load in mind, so the schematic column is not the best schematic result.

10 Critical path

10.1 Course hint vector

With $A = 1111$, $B = 0000$, toggling C_0 both ways gives both edges at each output. Rising C_0 alone gives only falling sums and rising C_4. All S outputs flip, as the hint predicts.

Table 12: Hint vector, extracted netlist

C0 edge	Output	Type	Delay (ps)
rise	S0	tpHL	72.8
rise	S1	tpHL	446.81
rise	S2	tpHL	402.78
rise	S3	tpHL	396.98
rise	C4	tpLH	300.03
fall	S0	tpLH	102.04
fall	S1	tpLH	454.15
fall	S2	tpLH	385.19
fall	S3	tpLH	385.81
fall	C4	tpHL	283.28

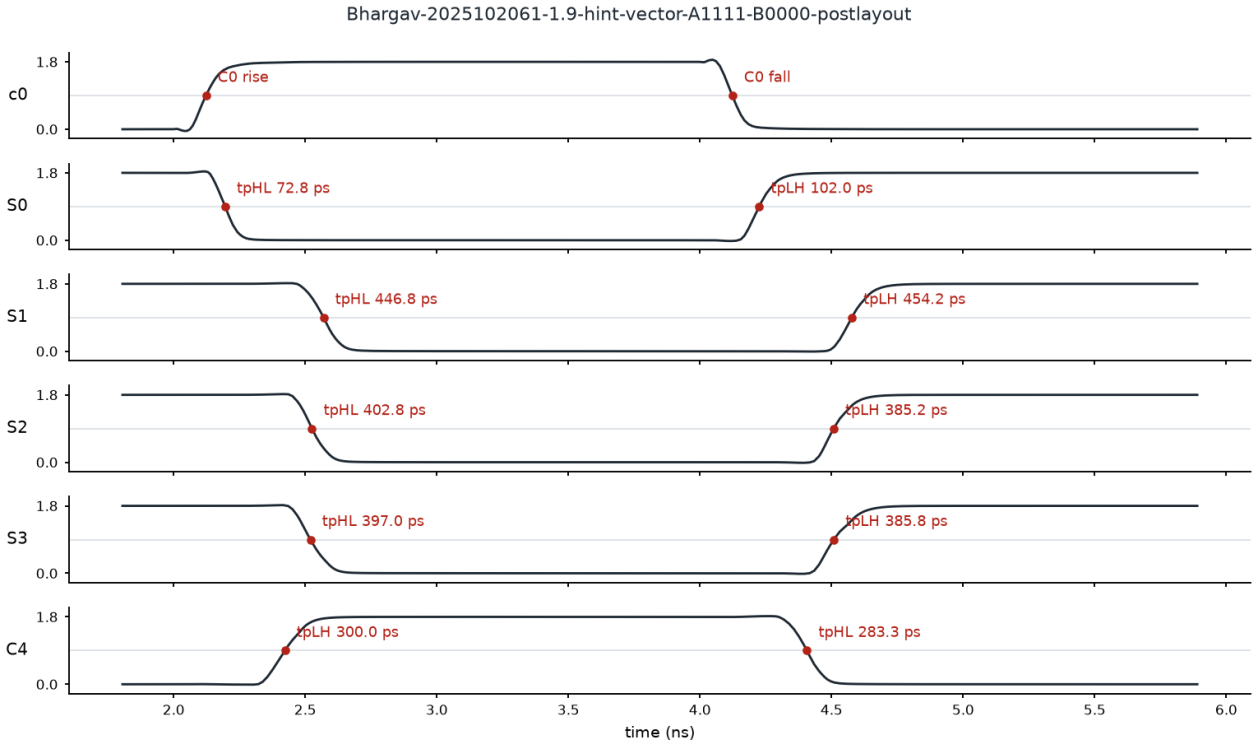


Figure 24: Hint vector, extracted netlist, with 50% crossings marked.

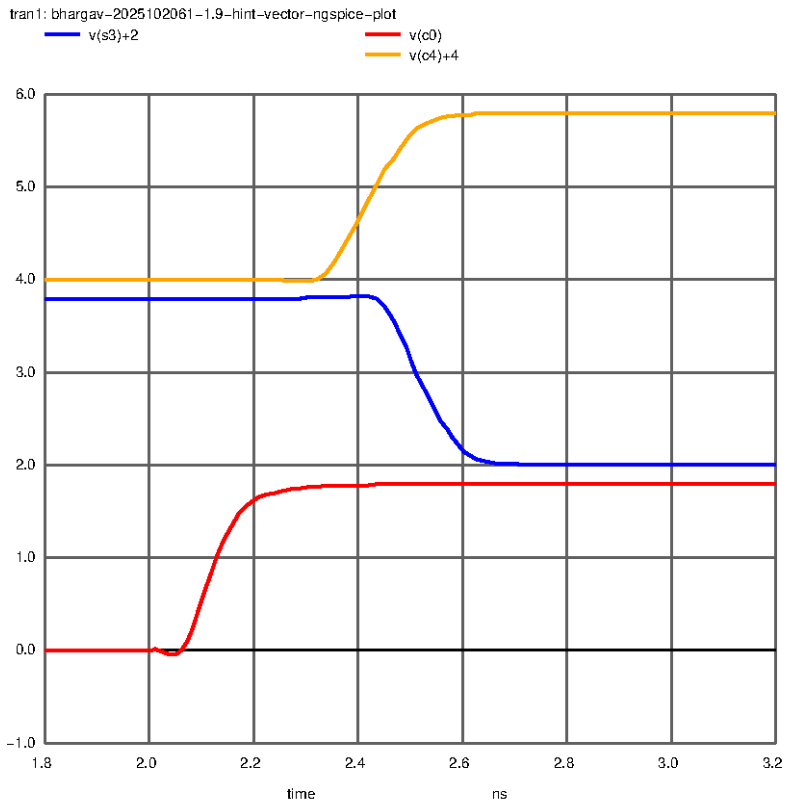


Figure 25: The same event as drawn by ngspice itself, showing the required curplottitle. ngspice displays the title in lower case. C0, S3+2 V, C4+4 V.

10.2 True worst case

The slowest hint-vector output is S1 at 454.1 ps, but the worst single-input transition is **a1 rise to s3 rise, A=0001 B=0101 C0=0** at **579.88 ps**. The hint vector skips the P/G stage and most of the prefix tree: C_0 reaches each carry through one inverter and one gate. Reporting only the hint vector would understate this design's delay by 135.7 ps.

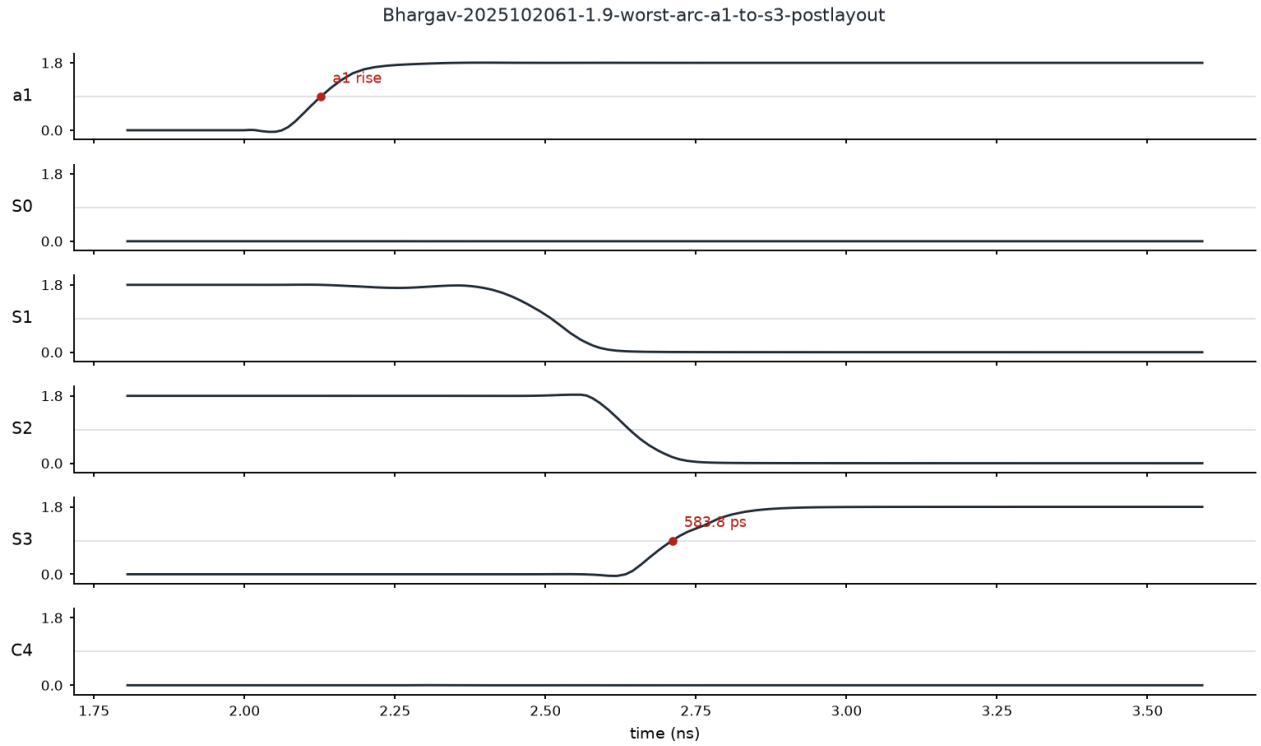


Figure 26: Worst single-input transition, extracted netlist.

History dependence. Delay depends slightly on what came before. Starting this arc from a settled DC state gives 583.8 ps, against 579.88 ps inside the long all-transition walk. For the hint vector, the walk measured up to 455.23 ps against 454.1 ps above. The likely cause is charge left on internal stack nodes, which was not investigated further. Differences of this size should be read as the precision of these numbers.

10.3 Several inputs changing together

Registers launch all inputs together, so several can change at once. All pairs would be 261,632 transitions, so an evenly spread sample was run: 65520 transitions (one segment in 4), on the schematic with extracted wire load annotated. Worst 577.64 ps, with 0 errors. The 100 slowest pairs from that sample were then re-simulated on the extracted netlist: worst **589.85 ps**, from 4 inputs changing, measured from the earliest of them. This exceeds the single-input worst case, so it is taken as $t_{pd,max}$. This is a sample, not an exhaustive proof, so an unsampled transition could be slightly slower.

11 Maximum clock frequency

The course defines $f_{max} = 1/(t_{pd,max} + t_{setup} + t_{hold})$. With post-layout values:

$$T = 589.85 + 38.0 + 45.7 = 673.6 \text{ ps}, \quad f_{max} = 1.4847 \text{ GHz}.$$

At the course-formula period (673.6 ps) the extracted registered circuit gave 264 wrong output samples, so that frequency is *not* reliable for this design. The formula leaves out clock-to-Q delay. A flip-flop launch actually needs $T \geq t_{cq} + t_{pd,max} + t_{setup} = 204.2 + 589.85 + 38.0 = 832.0$ ps, or 1.2019 GHz. Simulation of the extracted circuit finds the actual limit: **1.1467 GHz** (T = 872.1 ps). The value to rely on is the simulated limit.

12 Power

$P_{avg} = \frac{V_{DD}}{T_2 - T_1} \int_{T_1}^{T_2} I_{VDD}(t)\, dt$ for the extracted registered circuit, at the verified limit of 872.1 ps. Input drivers, clock buffers and output load inverters use a separate supply, so they are not included; the load's capacitance is still driven by the circuit. T_1 follows one settling cycle.

Table 13: Post-layout power, with PDP = Pavg x tpd,max

Activity	Cycles	Errors	Pavg (mW)	Energy/cycle (pJ)	PDP (fJ)
200 random plus critical vectors	264	0	3.3899	2.956	1999.51
Hint vector, C0 toggling each cycle	80	0	2.4954	2.176	1471.94
Inputs constant (clock and flip-flops)	40	0	0.8874	0.774	523.45

The first row is the main result. It is measured at the verified limit, not the course-formula frequency, because the circuit must run correctly for the number to mean anything.

13 Area-delay trade-off

ADP = area × delay, not to be confused with PDP. The final sizes were scaled uniformly, and each scale was laid out and extracted.

Table 14: Uniform scaling of the final sizes, post-layout

Scale	Width (lambda)	Worst single-input (ps)	Area (um2)	ADP (um2 ns)	P at 1 ns (mW)	PDP (fJ)
0.7	3252	602.38	2574.0	1550.5	0.9832	592.3
0.85	3857	601.39	2743.7	1650.0	1.1156	670.9
1.0	4536	579.88	2936.9	1703.0	1.2704	736.7
1.2	5456	590.99	3293.8	1946.6	1.4821	875.9

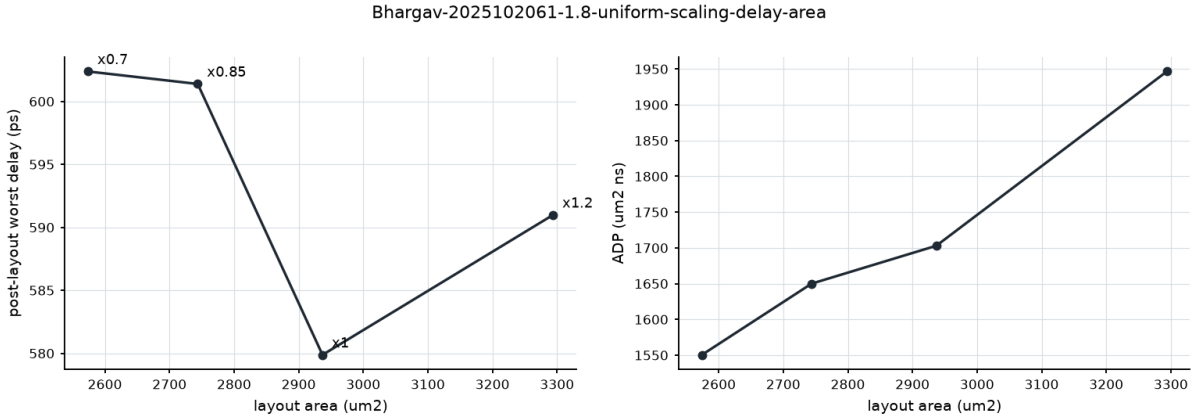


Figure 27: Delay and ADP against layout area, for uniform scaling.

Scale 1.0 is the chosen design, and repeating its layout from scratch reproduced the same delay. The curve is not smooth, because layout and routing change with each scale. Lower scales trade a small delay increase for less area and power, so they suit designs where ADP or PDP matters more than speed. Layouts at scales 1.3 and 1.4 were DRC clean but failed LVS after routing, so they gave no valid result and are not listed. The same assembly failure affected R with AOI22 sums. The final design passes LVS, but the cause of these failures was not found, so the router is not proven reliable at higher congestion.

14 Layout resistance

MAGIC's extraction here gives capacitance only, and its `extresist` command crashes on this layout (segmentation fault; log kept). As a substitute, the longest poly run from a contact to a gate end was measured from the layout: 152 λ, or 585.2 Ω at the technology file's 7.7 Ω/square. That resistance was placed in series with *every* one of the 194 gates, which is far more than any real gate sees, and all transitions were simulated again: worst single-input delay 579.88 to 600.83 ps, an increase of 20.95 ps (3.61%). Metal and vias are not included in this bound. Metal is 0.08 Ω/square and vias are 6 to 16 Ω, so they should matter much less than poly. This is an upper bound for poly only, not an extraction.

15 Limitations

- LVS is a project script (colour refinement on device-net graphs), not an industry tool. It compares devices and connectivity, and it has rejected injected faults and real routing shorts.
- Some larger layouts fail LVS after routing, for a cause not yet found, as noted in the trade-off section.
- Multi-input transitions are sampled, not exhaustive.
- Resistance is bounded for poly, not extracted. Reported delays exclude resistance.
- One typical-corner model at 27 C, with no voltage, temperature or process variation.
- The clock is ideal apart from one buffer; clock-tree layout and skew are not modelled beyond the extracted clock net.
- Figures of layouts are drawn from .mag data, since MAGIC's image export is blank in batch mode.
- R with AOI22 sums and C2 were not fully compared, as stated above, and R with TG sums did not get the layout-aware sizing round that C4 did. That round improved C4 by 4.8 to 17.7 ps, well short of the gap.
- Dynamic logic was not simulated. The planned domino prototype was not built. It was rejected on a timing argument: inputs launched at the rising edge would be non-monotonic during evaluation, and a half-cycle evaluate window conflicts with the course's delay measurement. This is reasoning, not a measurement.
- The simulated clock limit is about 40 ps longer than $t_{cq} + t_{pd,max} + t_{setup}$. This gap is not fully explained. Flip-flop values were measured in isolation, and the cause was not isolated further.
- TSPC minimum clock frequency was not characterised.

16 Netlists

All netlists and logs are in the project directory. The final adder and flip-flop netlists follow, with widths and lengths in metres and junctions as estimated before layout.

```
* nand2_x1p25: 2-input NAND
.subckt nand2_x1p25 i0 i1 y vdd gnd
Mp1 y i0 vdd vdd CMOSP W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Mp2 y i1 vdd vdd CMOSP W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Mn1 y i0 xn1 gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=6.0750e-13 PS=5.0400e-06 AD=1.2150e-12 PD=5.5800e-06
Mn2 xn1 i1 gnd gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=6.0750e-13 PD=5.0400e-06
.ends nand2_x1p25
* nor2_x1p56: 2-input NOR
.subckt nor2_x1p56 i0 i1 y vdd gnd
Mp1 xp1 i0 vdd vdd CMOSP W=5.5800e-06 L=1.8000e-07 AS=3.0132e-12 PS=1.2240e-05 AD=1.5066e-12 PD=1.1700e-05
Mp2 y i1 xp1 vdd CMOSP W=5.5800e-06 L=1.8000e-07 AS=1.5066e-12 PS=1.1700e-05 AD=3.0132e-12 PD=1.2240e-05
Mn1 y i0 gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn2 y i1 gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
.ends nor2_x1p56
* aoi21_x0p41: NOT(a b + c)
.subckt aoi21_x0p41 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mp2 xp1 b vdd vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mp3 y c xp1 vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn1 y a xn1 gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=1.9440e-13 PS=1.9800e-06 AD=3.8880e-13 PD=2.5200e-06
Mn2 xn1 b gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=1.9440e-13 PD=1.9800e-06
Mn3 y c gnd gnd CMOSN W=3.6000e-07 L=1.8000e-07 AS=1.9440e-13 PS=1.8000e-06 AD=1.9440e-13 PD=1.8000e-06
.ends aoi21_x0p41
* oai21_x0p41: NOT((a + b) c)
.subckt oai21_x0p41 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=3.8880e-13 PD=3.4200e-06
Mp2 y b xp1 vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=3.8880e-13 PS=3.4200e-06 AD=7.7760e-13 PD=3.9600e-06
Mp3 y c vdd vdd CMOSP W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
Mn1 y a xn1 gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
Mn2 y b xn1 gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
Mn3 xn1 c gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
.ends oai21_x0p41
* nor2_x1p95: 2-input NOR
.subckt nor2_x1p95 i0 i1 y vdd gnd
Mp1 xp1 i0 vdd vdd CMOSP W=7.0200e-06 L=1.8000e-07 AS=3.7908e-12 PS=1.5120e-05 AD=1.8954e-12 PD=1.4580e-05
Mp2 y i1 xp1 vdd CMOSP W=7.0200e-06 L=1.8000e-07 AS=1.8954e-12 PS=1.4580e-05 AD=3.7908e-12 PD=1.5120e-05
Mn1 y i0 gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mn2 y i1 gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
.ends nor2_x1p95
* nor2_x1p25: 2-input NOR
.subckt nor2_x1p25 i0 i1 y vdd gnd
Mp1 xp1 i0 vdd vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=2.4300e-12 PS=1.0080e-05 AD=1.2150e-12 PD=9.5400e-06
Mp2 y i1 xp1 vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=1.2150e-12 PS=9.5400e-06 AD=2.4300e-12 PD=1.0080e-05
Mn1 y i0 gnd gnd CMOSN W=1.0800e-06 L=1.8000e-07 AS=5.8320e-13 PS=3.2400e-06 AD=5.8320e-13 PD=3.2400e-06
Mn2 y i1 gnd gnd CMOSN W=1.0800e-06 L=1.8000e-07 AS=5.8320e-13 PS=3.2400e-06 AD=5.8320e-13 PD=3.2400e-06
.ends nor2_x1p25
* aoi21_x0p8: NOT(a b + c)
.subckt aoi21_x0p8 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Mp2 xp1 b vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Mp3 y c xp1 vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Mn1 y a xn1 gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=3.8880e-13 PS=3.4200e-06 AD=7.7760e-13 PD=3.9600e-06
Mn2 xn1 b gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=3.8880e-13 PD=3.4200e-06
Mn3 y c gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
.ends aoi21_x0p8
```

```
* oai21_x0p8: NOT((a + b) c)
.subckt oai21_x0p8 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=7.7760e-13 PD=6.3000e-06
Mp2 y b xp1 vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=7.7760e-13 PS=6.3000e-06 AD=1.5552e-12 PD=6.8400e-06
Mp3 y c vdd vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn1 y a xn1 gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn2 y b xn1 gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn3 xn1 c gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
.ends oai21_x0p8
* nand2_x1: 2-input NAND
.subckt nand2_x1 i0 i1 y vdd gnd
Mp1 y i0 vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mp2 y i1 vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mn1 y i0 xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Mn2 xn1 i1 gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
.ends nand2_x1
* aoi21_x1: NOT(a b + c)
.subckt aoi21_x1 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Mp2 xp1 b vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Mp3 y c xp1 vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Mn1 y a xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Mn2 xn1 b gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
Mn3 y c gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
.ends aoi21_x1
* oai21_x1: NOT((a + b) c)
.subckt oai21_x1 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=9.7200e-13 PD=7.7400e-06
Mp2 y b xp1 vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=9.7200e-13 PS=7.7400e-06 AD=1.9440e-12 PD=8.2800e-06
Mp3 y c vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mn1 y a xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mn2 y b xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mn3 xn1 c gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
.ends oai21_x1
* inv_x1: inverter
.subckt inv_x1 a y vdd gnd
Mp1 y a vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Mn1 y a gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
.ends inv_x1
* oai21_fn_x1p25: NOT((a + b) c)
.subckt oai21_fn_x1p25 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=2.4300e-12 PS=1.0080e-05 AD=1.2150e-12 PD=9.5400e-06
Mp2 y b xp1 vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=1.2150e-12 PS=9.5400e-06 AD=2.4300e-12 PD=1.0080e-05
Mp3 y c vdd vdd CMOSP W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Mn1 y c xn1 gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Mn2 xn1 a gnd gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Mn3 xn1 b gnd gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
.ends oai21_fn_x1p25
* nor2_x0p8: 2-input NOR
.subckt nor2_x0p8 i0 i1 y vdd gnd
Mp1 xp1 i0 vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=7.7760e-13 PD=6.3000e-06
Mp2 y i1 xp1 vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=7.7760e-13 PS=6.3000e-06 AD=1.5552e-12 PD=6.8400e-06
Mn1 y i0 gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
Mn2 y i1 gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
.ends nor2_x0p8
* inv_x1p25: inverter
.subckt inv_x1p25 a y vdd gnd
Mp1 y a vdd vdd CMOSP W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Mn1 y a gnd gnd CMOSN W=1.0800e-06 L=1.8000e-07 AS=5.8320e-13 PS=3.2400e-06 AD=5.8320e-13 PD=3.2400e-06
.ends inv_x1p25
* inv_x1p95: inverter
.subckt inv_x1p95 a y vdd gnd
Mp1 y a vdd vdd CMOSP W=3.5100e-06 L=1.8000e-07 AS=1.8954e-12 PS=8.1000e-06 AD=1.8954e-12 PD=8.1000e-06
Mn1 y a gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
.ends inv_x1p95
* aoi21_fnp_x0p41: NOT(a b + c)
.subckt aoi21_fnp_x0p41 a b c y vdd gnd
Mp1 xp1 c vdd vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mp2 y a xp1 vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mp3 y b xp1 vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn1 y b xn1 gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=1.9440e-13 PS=1.9800e-06 AD=3.8880e-13 PD=2.5200e-06
Mn2 xn1 a gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=1.9440e-13 PD=1.9800e-06
Mn3 y c gnd gnd CMOSN W=3.6000e-07 L=1.8000e-07 AS=1.9440e-13 PS=1.8000e-06 AD=1.9440e-13 PD=1.8000e-06
.ends aoi21_fnp_x0p41
* nand2_x0p41: 2-input NAND
.subckt nand2_x0p41 i0 i1 y vdd gnd
Mp1 y i0 vdd vdd CMOSP W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
Mp2 y i1 vdd vdd CMOSP W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=3.8880e-13 PD=2.5200e-06
Mn1 y i0 xn1 gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=1.9440e-13 PS=1.9800e-06 AD=3.8880e-13 PD=2.5200e-06
Mn2 xn1 i1 gnd gnd CMOSN W=7.2000e-07 L=1.8000e-07 AS=3.8880e-13 PS=2.5200e-06 AD=1.9440e-13 PD=1.9800e-06
.ends nand2_x0p41
* aoi21_fnp_x1: NOT(a b + c)
.subckt aoi21_fnp_x1 a b c y vdd gnd
Mp1 xp1 c vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Mp2 y a xp1 vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Mp3 y b xp1 vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Mn1 y b xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Mn2 xn1 a gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
Mn3 y c gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
.ends aoi21_fnp_x1
* inv_x1p56: inverter
.subckt inv_x1p56 a y vdd gnd
Mp1 y a vdd vdd CMOSP W=2.7900e-06 L=1.8000e-07 AS=1.5066e-12 PS=6.6600e-06 AD=1.5066e-12 PD=6.6600e-06
Mn1 y a gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
.ends inv_x1p56
* oai21_x0p51: NOT((a + b) c)
.subckt oai21_x0p51 a b c y vdd gnd
Mp1 xp1 a vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
Mp2 y b xp1 vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Mp3 y c vdd vdd CMOSP W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
Mn1 y a xn1 gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
Mn2 y b xn1 gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
Mn3 xn1 c gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
.ends oai21_x0p51
* oai21_fnp_x0p8: NOT((a + b) c)
.subckt oai21_fnp_x0p8 a b c y vdd gnd
Mp1 xp1 b vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=7.7760e-13 PD=6.3000e-06
Mp2 y a xp1 vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=7.7760e-13 PS=6.3000e-06 AD=1.5552e-12 PD=6.8400e-06
Mp3 y c vdd vdd CMOSP W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn1 y c xn1 gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn2 xn1 a gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
Mn3 xn1 b gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=7.7760e-13 PD=3.9600e-06
.ends oai21_fnp_x0p8
* xor_aoi22_x1p25_0p51: dual-rail XOR, AOI22 with c-bar inverter
.subckt xor_aoi22_x1p25_0p51 p pb c s vdd gnd
Ms0p1 cb c vdd vdd CMOSP W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
Ms0n1 cb c gnd gnd CMOSN W=4.5000e-07 L=1.8000e-07 AS=2.4300e-13 PS=1.9800e-06 AD=2.4300e-13 PD=1.9800e-06
Ms1p1 s1xp1 p vdd vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=2.4300e-12 PS=1.0080e-05 AD=2.4300e-12 PD=1.0080e-05
Ms1p2 s1xp1 c vdd vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=2.4300e-12 PS=1.0080e-05 AD=2.4300e-12 PD=1.0080e-05
Ms1p3 s pb s1xp1 vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=2.4300e-12 PS=1.0080e-05 AD=2.4300e-12 PD=1.0080e-05
Ms1p4 s cb s1xp1 vdd CMOSP W=4.5000e-06 L=1.8000e-07 AS=2.4300e-12 PS=1.0080e-05 AD=2.4300e-12 PD=1.0080e-05
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Ms1n1 s p s1xn1 gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=6.0750e-13 PS=5.0400e-06 AD=1.2150e-12 PD=5.5800e-06
Ms1n2 s1xn1 c gnd gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=6.0750e-13 PD=5.0400e-06
Ms1n3 s pb s1xn2 gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=6.0750e-13 PS=5.0400e-06 AD=1.2150e-12 PD=5.5800e-06
Ms1n4 s1xn2 cb gnd gnd CMOSN W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=6.0750e-13 PD=5.0400e-06
.ends xor_aoi22_x1p25_0p51
* xor_aoi22_x1_0p64: dual-rail XOR, AOI22 with c-bar inverter
.subckt xor_aoi22_x1_0p64 p pb c s vdd gnd
Ms0p1 cb c vdd vdd CMOSP W=1.1700e-06 L=1.8000e-07 AS=6.3180e-13 PS=3.4200e-06 AD=6.3180e-13 PD=3.4200e-06
Ms0n1 cb c gnd gnd CMOSN W=5.4000e-07 L=1.8000e-07 AS=2.9160e-13 PS=2.1600e-06 AD=2.9160e-13 PD=2.1600e-06
Ms1p1 s1xp1 p vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Ms1p2 s1xp1 c vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Ms1p3 s pb s1xp1 vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Ms1p4 s cb s1xp1 vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Ms1n1 s p s1xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1n2 s1xn1 c gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
Ms1n3 s pb s1xn2 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1n4 s1xn2 cb gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
.ends xor_aoi22_x1_0p64
* xor_aoi22_x0p8_1p25: dual-rail XOR, AOI22 with c-bar inverter
.subckt xor_aoi22_x0p8_1p25 p pb c s vdd gnd
Ms0p1 cb c vdd vdd CMOSP W=2.2500e-06 L=1.8000e-07 AS=1.2150e-12 PS=5.5800e-06 AD=1.2150e-12 PD=5.5800e-06
Ms0n1 cb c gnd gnd CMOSN W=1.0800e-06 L=1.8000e-07 AS=5.8320e-13 PS=3.2400e-06 AD=5.8320e-13 PD=3.2400e-06
Ms1p1 s1xp1 p vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Ms1p2 s1xp1 c vdd vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Ms1p3 s pb s1xp1 vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Ms1p4 s cb s1xp1 vdd CMOSP W=2.8800e-06 L=1.8000e-07 AS=1.5552e-12 PS=6.8400e-06 AD=1.5552e-12 PD=6.8400e-06
Ms1n1 s p s1xn1 gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=3.8880e-13 PS=3.4200e-06 AD=7.7760e-13 PD=3.9600e-06
Ms1n2 s1xn1 c gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=3.8880e-13 PD=3.4200e-06
Ms1n3 s pb s1xn2 gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=3.8880e-13 PS=3.4200e-06 AD=7.7760e-13 PD=3.9600e-06
Ms1n4 s1xn2 cb gnd gnd CMOSN W=1.4400e-06 L=1.8000e-07 AS=7.7760e-13 PS=3.9600e-06 AD=3.8880e-13 PD=3.4200e-06
.ends xor_aoi22_x0p8_1p25
* xor_aoi22_x0p51_1: dual-rail XOR, AOI22 with c-bar inverter
.subckt xor_aoi22_x0p51_1 p pb c s vdd gnd
Ms0p1 cb c vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms0n1 cb c gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=4.8600e-13 PD=2.8800e-06
Ms1p1 s1xp1 p vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1p2 s1xp1 c vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1p3 s pb s1xp1 vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1p4 s cb s1xp1 vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1n1 s p s1xn1 gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=2.4300e-13 PS=2.3400e-06 AD=4.8600e-13 PD=2.8800e-06
Ms1n2 s1xn1 c gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=2.4300e-13 PD=2.3400e-06
Ms1n3 s pb s1xn2 gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=2.4300e-13 PS=2.3400e-06 AD=4.8600e-13 PD=2.8800e-06
Ms1n4 s1xn2 cb gnd gnd CMOSN W=9.0000e-07 L=1.8000e-07 AS=4.8600e-13 PS=2.8800e-06 AD=2.4300e-13 PD=2.3400e-06
.ends xor_aoi22_x0p51_1
* cla_final: Candidate C4: valency-2 prefix, late C0
.subckt cla_final a0 a1 a2 a3 b0 b1 b2 b3 c0 s0 s1 s2 s3 c4 vdd gnd
Xnand0 a0 b0 g0b vdd gnd nand2_x1p25
Xnor0 a0 b0 t0b vdd gnd nor2_x1p56
Xaoi0 a0 b0 t0b p0 vdd gnd aoi21_x0p41
Xoai0 a0 b0 g0b p0b vdd gnd oai21_x0p41
Xnand1 a1 b1 g1b vdd gnd nand2_x1p25
Xnor1 a1 b1 t1b vdd gnd nor2_x1p95
Xaoi1 a1 b1 t1b p1 vdd gnd aoi21_x0p41
Xoai1 a1 b1 g1b p1b vdd gnd oai21_x0p41
Xnand2 a2 b2 g2b vdd gnd nand2_x1p25
Xnor2 a2 b2 t2b vdd gnd nor2_x1p25
Xaoi2 a2 b2 t2b p2 vdd gnd aoi21_x0p8
Xoai2 a2 b2 g2b p2b vdd gnd oai21_x0p8
Xnand3 a3 b3 g3b vdd gnd nand2_x1
Xnor3 a3 b3 t3b vdd gnd nor2_x1p25
Xaoi3 a3 b3 t3b p3 vdd gnd aoi21_x1
Xoai3 a3 b3 g3b p3b vdd gnd oai21_x1
Xinvc0 c0 c0b vdd gnd inv_x1
Xg10 t1b g0b g1b g1_0 vdd gnd oai21_fn_x1p25
Xt10 t1b t0b t1_0 vdd gnd nor2_x1p56
Xg32 t3b g2b g3b g3_2 vdd gnd oai21_x0p8
Xt32 t3b t2b t3_2 vdd gnd nor2_x0p8
Xig2 g2b g2 vdd gnd inv_x1p25
Xit2 t2b t2 vdd gnd inv_x1p95
Xg30 t3_2 g1_0 g3_2 g3_0b vdd gnd aoi21_fnp_x0p41
Xt30 t3_2 t1_0 t3_0b vdd gnd nand2_x0p41
Xg20 t2 g1_0 g2 g2_0b vdd gnd aoi21_fnp_x1
Xt20 t2 t1_0 t2_0b vdd gnd nand2_x1p25
Xig10 g1_0 g1_0b vdd gnd inv_x1
Xit10 t1_0 t1_0b vdd gnd inv_x1p56
Xc1 t0b c0b g0b c1 vdd gnd oai21_x0p51
Xc2 t1_0b c0b g1_0b c2 vdd gnd oai21_fn_x1p25
Xc3 t2_0b c0b g2_0b c3 vdd gnd oai21_fn_x1p25
Xc4g t3_0b c0b g3_0b c4 vdd gnd oai21_fnp_x0p8
Xx0 p0 p0b c0 s0 vdd gnd xor_aoi22_x1p25_0p51
Xx1 p1 p1b c1 s1 vdd gnd xor_aoi22_x1_0p64
Xx2 p2 p2b c2 s2 vdd gnd xor_aoi22_x0p8_1p25
Xx3 p3 p3b c3 s3 vdd gnd xor_aoi22_x0p51_1
.ends cla_final

* dff_tspc_x0p6_1_1_2: TSPC positive-edge flip-flop
.subckt dff_tspc_x0p6_1_1_2 d clk q vdd gnd
Ms0p1 s0xp1 d vdd vdd CMOSP W=2.1600e-06 L=1.8000e-07 AS=1.1664e-12 PS=5.4000e-06 AD=5.8320e-13 PD=4.8600e-06
Ms0p2 a clk s0xp1 vdd CMOSP W=2.1600e-06 L=1.8000e-07 AS=5.8320e-13 PS=4.8600e-06 AD=1.1664e-12 PD=5.4000e-06
Ms0n1 a d gnd gnd CMOSN W=5.4000e-07 L=1.8000e-07 AS=2.9160e-13 PS=2.1600e-06 AD=2.9160e-13 PD=2.1600e-06
Ms1p1 b clk vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1n1 b a s1xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Ms1n2 s1xn1 clk gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
Ms2p1 qb b vdd vdd CMOSP W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
Ms2n1 qb clk s2xn1 gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=4.8600e-13 PS=4.1400e-06 AD=9.7200e-13 PD=4.6800e-06
Ms2n2 s2xn1 b gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=4.8600e-13 PD=4.1400e-06
Ms3p1 q qb vdd vdd CMOSP W=3.6000e-06 L=1.8000e-07 AS=1.9440e-12 PS=8.2800e-06 AD=1.9440e-12 PD=8.2800e-06
Ms3n1 q qb gnd gnd CMOSN W=1.8000e-06 L=1.8000e-07 AS=9.7200e-13 PS=4.6800e-06 AD=9.7200e-13 PD=4.6800e-06
.ends dff_tspc_x0p6_1_1_2
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